

Slack Variables for Non-Linearly Separable Case (Soft Margin SVM)

Motivation:

In real-world datasets, data points may not be perfectly linearly separable. Some points may overlap, be noisy, or be outliers.

To handle such situations, SVM uses slack variables to allow some violations of the margin.

Slack Variables (ξ_i):

Slack variables $\xi_i \geq 0$ are introduced to relax the hard margin constraint.

Without slack:

$$y_i (w^T x_i + b) \geq 1$$

With slack:

$$y_i (w^T x_i + b) \geq 1 - \xi_i, \text{ where } \xi_i \geq 0$$

Interpretation:

- $\xi_i = 0$: Point is correctly classified and outside margin
- $0 < \xi_i < 1$: Point is inside the margin but correctly classified
- $\xi_i > 1$: Point is misclassified

Optimization Objective:

$$\text{Minimize } (1/2) * ||w||^2 + C * \sum(\xi_i)$$

$$\text{Subject to: } y_i (w^T x_i + b) \geq 1 - \xi_i \text{ and } \xi_i \geq 0$$

Role of C (Penalty Parameter):

- High C: Less tolerance for violations -> hard margin -> risk of overfitting
- Low C: More tolerance for violations -> soft margin -> better generalization

Summary Table:

$\xi_i = 0$	-> Correctly classified, outside margin
$0 < \xi_i < 1$	-> Correctly classified, inside margin
$\xi_i > 1$	-> Misclassified
C	-> Controls trade-off between margin size and classification errors

These slack variables allow the SVM to balance margin maximization with real-world tolerance for noise and overlap.